Mambo

Track: Development — AI for Impact Challenge (AI4I 2026)

Project Title: Mambo — a whole-of-government-ready citizen information assistant

Team Name: Team Mambo

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Live demo: https://mambo.yttrix.tech (citizen chat) · https://mambo.yttrix.tech/admin (ministry portal)

1. Problem Definition & Strategic Alignment

The problem. Zimbabwean citizens struggle to find reliable, plain-language government information. Official content is scattered across many ministry websites, frequently as scanned PDFs, written in legal jargon, and rarely answers the *whole* question — what to bring, what it costs, where to go. Citizens do not know which ministry handles what, so they make avoidable trips to the wrong offices. The cost is wasted time, uneven access, and frustration — sharpest for citizens on low-bandwidth mobile connections and for front-line registry staff who answer the same questions all day.

Who it serves. Citizens applying for IDs, passports, certificates and tax clearance; SME owners navigating business and tax registration; students and journalists verifying laws and policies; and ministry helpdesk/registry staff who need consistent, cited, first-line answers. A mobile-first, free, always-on assistant reaches the widest audience.

What Mambo is. A retrieval-augmented assistant that answers in plain language using **only** the retrieved documents from an allow-listed corpus of ministry, agency, tax, education, and public legal sources, with citations, an honest **evidence-status badge** on every answer, structured **service-journey cards** for common tasks, and a **ministry handoff card** when it cannot answer. *Mambo* is a Shona word for “king”; here it is framed as the **king of information**.

Strategic alignment. Mambo implements the National AI Strategy’s goals for inclusive, trustworthy public-service AI built on local capability and digital sovereignty: allow-listed-source grounding with citations (trustworthy); abstention and honest status (responsible); mobile-first free access (inclusive); a self-hostable open-source stack with a CCE roadmap (sovereign, local). It is whole-of-government-ready: led from the ICT use case, currently covering 5 ministries plus 4 adjacent sources, and designed so additional ministries plug in via a curated registry. The product includes a **customer-service portal** for covered ministries to see what citizens ask, curate vetted answers, and handle handoffs.

2. Technical Design & Product Logic

Architecture — three modules on different rhythms, meeting at one Knowledge Store. Splitting by rhythm lets each scale independently:

```
 ingestion/ (batch)  ── discover · OCR · chunk · embed ──▶  PostgreSQL + pgvector
                                                             (Knowledge Store)
 webchat/  ──/api──▶  rag/ (FastAPI, per question)
──route·retrieve·generate·trust──▶
 (Next.js 14)                                                       reads ▲
```

Per-question pipeline: `guard → intent → route → retrieve → (confident?) → generate or fall back → trust layer`. The safety guard abstains before retrieval (no LLM needed); generation sits behind a **swappable** OpenAI-compatible interface.

Models and stack.

Component	Choice	Why
Generation	DeepSeek (flash), swappable	Phase 2 self-host an open-weight model on the CCE — no code change

Component	Choice	Why
Embeddings	Qwen3-Embedding-8B (4096-dim), self-hosted (Ollama)	Multilingual → Shona/Ndebele is an addition, not a re-embed
Knowledge store	PostgreSQL 16 + pgvector	Runs on a cheap cloud box *and* a Ministry server — one tech, two homes
Retrieval	Ministry-scoped cosine + recency boost	Newest version of an amended law wins
Webchat	Next.js 14, streaming SSE, mobile-first	Low-bandwidth, always-on

Data. 890 documents / 2,901 chunks from 9 enabled sources (5 ministries + ZIMRA, ZIMSEC, Veritas, ZimLII; selected embassy/consulate pages are included under Home Affairs where listed in the registry allow-list). Every chunk stores source URL, page, fetch date, content hash and raw path — so every citation is verifiable.

Why AI is necessary (not a sledgehammer). Plain-language synthesis, multi- ministry routing, conversational follow-ups ("what about for schools?"), and cited answers over many long documents cannot be met by keyword search, static FAQs, or SQL. RAG is the appropriate, minimal use of AI: retrieve official passages and ground generation in them. We do not apply AI where a rule would do.

Product features (built). Ministry routing ("find the right office"); inline `[n]` citations on every answer; honest **evidence-status badge** (answered / partial / unsupported / declined); a deterministic **abstention guard** (medical, legal, personal-data, political, prompt-injection); **service-journey cards** for six common tasks; **ministry handoff cards** plus a persistent **"Talk to a human"** flow — covered-source contacts (call / WhatsApp / email / office hours, where available in the registry) one tap away at any time, not only when an answer falls back; switching ministry focus starts a **fresh conversation** (no cross-ministry context bleed); **live "thinking" steps** that narrate the retrieval; streaming; mobile-first; light/dark themes; a **designed national identity** (Spectral/Hanken typography, flag-palette tokens, a Seal/Mark/Wordmark); and a **ministry customer-service portal** (`/admin`) with per-ministry login, a scoped analytics dashboard (top questions, fallback rate, feedback), and **reviewed-answer curation** — staff vet the top questions, and citizens get those answers **instantly** (sub-second, zero LLM cost) via the reviewed-answer short-circuit.

Measured results (not asserted). A fresh build from a clean checkout boots; **67 automated tests pass**. On the corpus and a 32-question citizen-query set:

Metric	Result
Router accuracy (English)	89.3% routed to an expected ministry
Dangerous-case abstention	7 / 7 handled correctly
Corpus integrity checks	8 / 8 pass
Citation link liveness (sample)	8 / 8 resolved
Allow-listed-source-only retrieval	True (web verifier off by default)
Reviewed-answer latency	<1 s in the reviewed-answer path (zero LLM call)

Honest gaps are disclosed, not hidden: education (3 chunks) and ZimLII (1) are thin; Shona/Ndebele router accuracy is 60% (Phase 2; embeddings are already multilingual).

3. Deliverables & CCE Implementation Roadmap

What is delivered. A working **two-sided** product: a citizen-facing RAG assistant (ingestion → retrieve → generate → trust) AND a ministry customer-service portal (per-ministry login, analytics dashboard, reviewed-answer curation). Both are deployed and demonstrable; the codebase includes 67 tests, a full evidence pack, and a designed national identity system.

Timeline (8 weeks of AI4I support).

Weeks	Workstream	Output
1–2	Harden	Close remaining security/schema/test items; run full RAG eval on resourced compute; fix accessibility “pending” items
3–4	Expand	Close coverage gaps (re-crawl education; expand ZimLII); add Shona/Ndebele registry keywords; measured sn/nd pilot
5–8	Pilot	Home Affairs pilot: confirm sources + handoff contacts, reviewed-answer cache, controlled user group, weekly review, go/no-go

ZCHPC CCE roadmap. The stack is CCE-ready by design (open-source, self-hostable, swappable model interface). On the CCE we run PostgreSQL+pgvector (the Knowledge Store), bulk embedding jobs on GPU (`ingestion/embed_bulk.py` already targets a remote GPU Ollama endpoint), a self-hosted open-weight generation model behind the swappable interface, and the evaluation pipeline. Migration is a re-point of `OLLAMA_BASE_URL` / `DEEPSEEK_BASE_URL` — no code change.

CCE milestone	Outcome
Provision	CCE partition: Postgres+pgvector + Ollama/vLLM endpoints
Migrate	Move the Knowledge Store to CCE (dump/restore; same schema)
Re-embed	Bulk embeddings on GPU; run full evaluation
Benchmark	Latency/cost vs current local+API setup; publish metrics
Operate	Incremental refresh + monitoring from CCE

This removes per-query API cost and keeps data in-country. **Mambo is cloud-deployed today, not edge**; the CCE is the documented scale/sovereignty path, and mobile-first low-bandwidth is the user-facing strategy.

4. Compliance & Risk Mitigation

Data Protection Act [Chapter 12:07].

Obligation	How Mambo handles it
Lawful basis	Public official documents (low personal-data risk)
Consent	**Data-use consent checkbox** before the first question: asking is blocked until it is ticked; the choice is stored client-side (versioned key, timestamped) and is revocable by unticking — no account or server-side identity is created
Data minimisation	Query log keeps only needed fields; `client_ip`/`user_agent` retention-bounded
Retention & deletion	Defined retention window + deletion workflow (planned PII redaction)
Access control	Restricted log access by role
Transparency	Inline notice beside the consent checkbox: do not enter ID numbers, phone numbers, or medical information
Open-data release	**Not** claimed for query logs

Cybersecurity. Nonce challenge (single-use) on write endpoints; origin lock to the official app; per-IP rate limiting + concurrent-stream cap; bot user-agent blocklist + behaviour/entropy checks; validated input sizes; docs/redoc disabled in production; secrets never committed. The deterministic abstention guard and allow-listed-source retrieval hardens the content layer. (67 tests cover nonce, abstention, security, schema, and the reviewed-answer streaming path.) The admin portal adds **session-cookie auth** (bcrypt, httpOnly, SameSite-Lax, ministry-scoped SQL so staff only see their own data; rate-limited login; generic 401).

Responsible AI. Mandatory citations; a confidence threshold below which Mambo does not guess; the evidence-status badge; abstention for out-of-scope/unsafe questions with a referral (clinician / Law Society / ministry); per-section "not specified" in journeys; recency boost for amended laws; a risk register and monitoring plan with drift detection (PSI/KL) and a rule-baseline fallback.

Key risks (full register in `submission/deployment/risk_register.csv`): hallucination (grounding + evidence status + abstention); coverage gaps (disclosed, being closed); latency on local hosting (CCE roadmap); local-language inequity (honestly disclosed, Phase 2); PII in queries (minimisation + redaction plan).

5. Sustainability & Future Adoption

Cost reality. Infrastructure is inexpensive (~USD 50/month at pilot); the real cost is people — a reviewer per ministry batch and part-time engineering (`submission/business/cost_model.csv`). The stack is open-source and self-hostable, so a Ministry can run it on its own servers with **no per-query API cost** once on the CCE.

Adoption path. Pilot with one ministry (Home Affairs proposed) under AI4I milestone support → per-ministry onboarding (repeatable, via `ministry_onboarding_pack.md`) → whole-of-government rollout (the registry is the keystone) → embeddable widget on live ministry sites + low-bandwidth channels (WhatsApp).

Maintainability. Incremental refresh re-ingests only changed documents; a quarterly re-verification cadence; the **reviewed-answer cache** — now live in the admin portal, where ministry staff curate the top questions and citizens get those answers instantly (sub-second, no LLM call); monitoring that drives what to fix next.

A note on fallibility. Mambo is a **guidance and routing assistant**, not a replacement for official decisions, professional advice, or ministry confirmation. It points citizens to the right office and the right documents, and it says so honestly when the official record does not cover a question. That scope discipline is how it earns trust.